
Predicting Interference in Model Merging

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Leonardo Lavezzari

Abstract

Model merging aims to combine independently fine-tuned models without joint retraining, but the result can lose part of the specialist performance when task updates disagree. We study whether the geometry of task vectors can anticipate this interference. Starting from a common compact BERT encoder, we train specialists for six binary NLP tasks and merge their encoders pairwise while retaining the original task heads. Across multiple seeds, related and better-aligned updates are generally easier to merge. Longer training improves every specialist but increases average merge damage. TIES-style coordinate conflict resolution is more reliable than uniform averaging. A final held-out experiment shows that converting directional diagnostics into one scalar weight per task does not improve merging, indicating that interference must be handled at a finer granularity.

1. Introduction

Fine-tuned models derived from the same initialization can be combined directly in weight space, avoiding the inference and storage cost of running several complete models. Simple weight averaging can work when solutions occupy a compatible region (Wortsman et al., 2022), while Task Arithmetic represents fine-tuning as an update from a shared base model (Ilharco et al., 2023). However, adding independently learned updates can produce destructive interference. TIES addresses this problem by trimming small updates, resolving sign conflicts, and merging only the elected coordinates (Yadav et al., 2023).

This work asks a narrower question: *can task-vector geometry predict which specialists are difficult to merge?* We first test whether cosine alignment and sign agreement are associated with retained performance across task pairs and

seeds. We then localize interference across encoder layers, extend the study to more task families and longer training, and evaluate a directional reweighting rule on held-out seeds. The objective is not to obtain state-of-the-art benchmark scores, but to explain when controlled pairwise merging succeeds or fails.

2. Method

Let θ_0 denote a pretrained encoder and θ_i the encoder fine-tuned for task i . Its task vector is $\tau_i = \theta_i - \theta_0$. For a pair (i, j) , Mean uses $\theta_0 + (\tau_i + \tau_j)/2$, while Task Arithmetic uses $\theta_0 + \lambda(\tau_i + \tau_j)$ with the frozen value $\lambda = 0.75$. We evaluate both the tensor-wise implementation of TIES and the flatten-first global trimming scope, with density 0.2 and scale 1.0. Only encoder parameters are merged; each task keeps the classification head learned by its specialist.

Compatibility is described by cosine similarity and coordinate sign agreement. To expose asymmetry hidden by a pair average, we also compute the projection of an incoming update onto the source task vector and their norm ratio. Finally, projection-balanced merging converts these directional quantities into complementary pair-level weights. Seed 42 is used for development and seeds 7 and 123 for its held-out evaluation; the weights are fixed before observing held-out results.

3. Experimental Setup

All specialists start from `prajjwall/bert-mini`, a compact BERT model (Devlin et al., 2019; Turc et al., 2019). The tasks cover sentence sentiment (SST-2), document sentiment (IMDb), paraphrase detection (MRPC), textual entailment (RTE), linguistic acceptability (CoLA), and binary question answering (BoolQ). SST-2, MRPC, RTE, and CoLA come from GLUE (Wang et al., 2018); IMDb follows (Maas et al., 2011). Accuracy is used except for MRPC (F1) and CoLA (Matthews correlation).

Training uses at most 12,000 examples and evaluation at most 2,000, with fixed subsets, sequence length 128, batch size 32, learning rate 2×10^{-5} , and seeds 7, 42, and 123. The main specialists are trained for 1,200 optimizer steps. A controlled seed-42 ablation compares 400 and 1,200

Email: Leonardo Lavezzari <lavez-zari.1984079@studenti.uniroma1.it>.

Table 1. Held-out 1,200-step results on 20 pair-seed units excluding CoLA. Values are task-score changes from the corresponding specialists; retention is the merged-to-specialist score ratio.

Method	Mean Δ	SD	Worst	Ret.
Tensor-wise TIES	-0.0534	0.0276	-0.1435	0.926
Global TIES	-0.0615	0.0314	-0.1794	0.915
Projection-balanced	-0.0624	0.0317	-0.2255	0.915
Mean	-0.0631	0.0308	-0.2265	0.914
Task Arithmetic	-0.0645	0.0407	-0.2000	0.911

steps on identical subsets. Performance change is always measured against the corresponding specialist, so $\Delta = 0$ means no loss after merging.

4. Results and Discussion

In the initial four-task confirmation, cosine similarity is positively associated with retained Mean-merge performance in every seed (Pearson 0.718, 0.520, and 0.878). The average correlation is 0.705, whereas raw sign conflict is weaker. Related sentiment specialists are among the easiest pairs, while pairs containing RTE are consistently fragile. Layer interventions show that embeddings and late blocks can be shared almost without loss, while sharing all Transformer blocks causes a much larger degradation. Interference is therefore cumulative and distributed rather than isolated in one final layer.

Increasing training from 400 to 1,200 steps improves every seed-42 specialist, including CoLA from 0.000 to 0.110 Matthews correlation. Nevertheless, average merge degradation becomes more negative for every frozen method: -0.0336 for Mean, -0.0252 for Task Arithmetic, -0.0245 for tensor-wise TIES, and -0.0297 for global TIES. Stronger task-specific updates are therefore not automatically more compatible.

Table 1 reports the pre-declared held-out comparison on long-trained specialists. Tensor-wise TIES gives the best mean change, retention, and worst observed loss. Projection-balanced merging is worse than tensor-wise TIES by -0.00894 on average, wins only 25% of the 20 paired cases, and has a bootstrap 95% interval $[-0.01485, -0.00304]$. Its learned weights remain close to 0.5 (range 0.479–0.533), making it behave similarly to Mean. Directional diagnostics still replicate on held-out seeds, but compressing them into one scalar per task discards the local sign conflicts that TIES preserves.

5. Conclusion

Task-vector geometry provides useful, though not universal, evidence about merge compatibility. The experiments consistently favor conflict-aware merging over uni-

form averaging and show that better specialists may become harder to combine. The negative held-out result is also informative: pair-level scalar balancing is too coarse, and future methods should act on conflicting coordinates or parameter groups. Code and complete result tables are available at <https://github.com/LeuxLello/Dlai-model-merging>.

6. Statement on the Use of AI

Throughout the project, extensive use was made of AI tools, which helped to refine the research question, design the experimental sequence, and implement the code for training, evaluation, fusion and diagnostics of the models. They also assisted in identifying scientific references to further refine the final work.

References

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Appendix

Protocol and scope. The complete study contains eleven executable notebooks. The final six-task protocol uses 15 task pairs per seed. The 400-step extension contains 45 pair-seed units and 90 oriented task observations. The long-training held-out comparison uses ten no-CoLA pairs on each of seeds 7 and 123. CoLA is excluded from this primary comparison because it is the weakest specialist and failed to learn at 400 steps; complete tables including CoLA remain in the repository.

Additional controls. Equalizing task-vector norms does not improve Mean for any four-task pair, weakening a simple magnitude-only explanation. Development-selected early-layer attenuation chooses the unmodified baseline, as does the structured TIES density experiment. These negative controls reduce the chance that the main result follows from one easily corrected scale or layer effect.

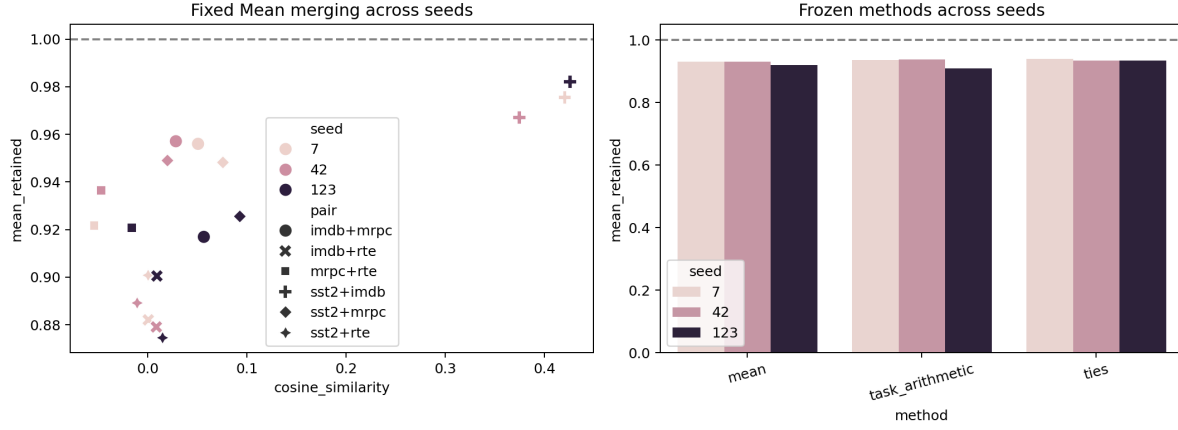


Figure 1. Multi-seed confirmation. Cosine alignment is positively associated with retained Mean-merge performance in all three seeds; frozen TIES configurations are more stable overall.

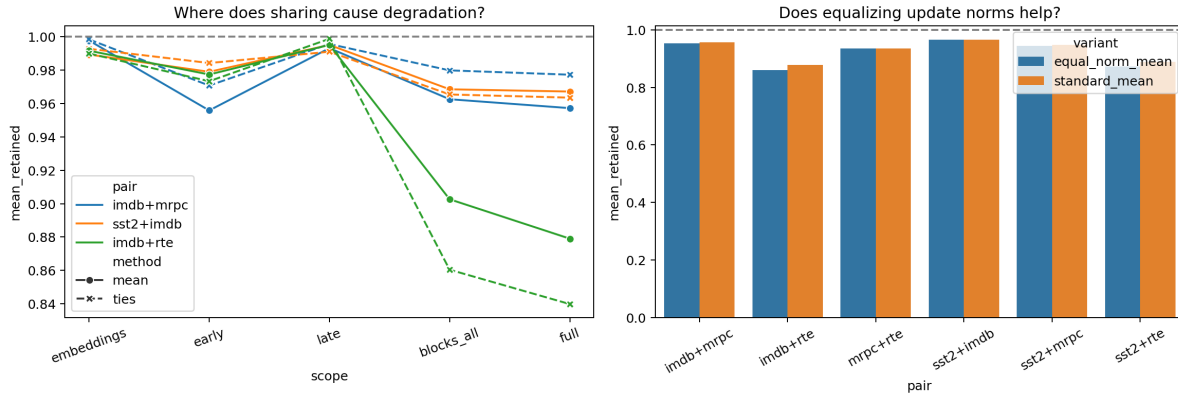


Figure 2. Layer-wise interventions and equal-norm control. The degradation becomes substantial when several Transformer blocks are shared together rather than from one isolated late scope.

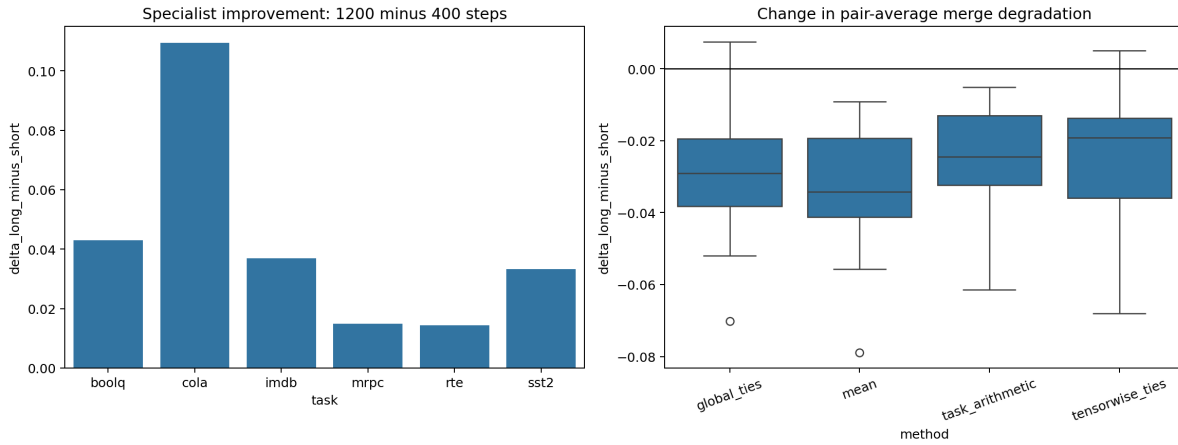


Figure 3. Training-budget ablation. Longer training improves all seed-42 specialists but increases average merging damage for every frozen method.

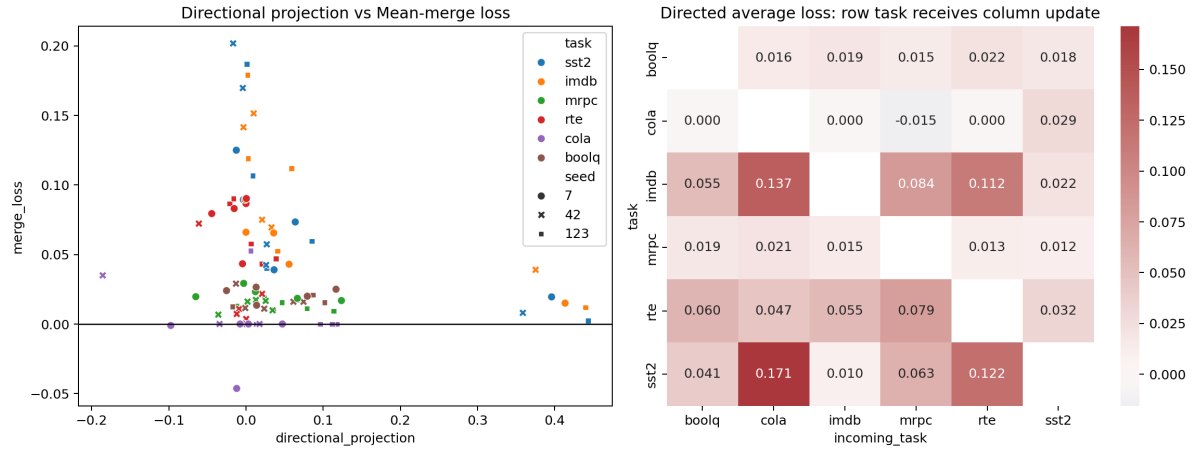


Figure 4. Directional diagnostics reveal that the two tasks in a pair can experience different losses even when they share the same merged encoder.

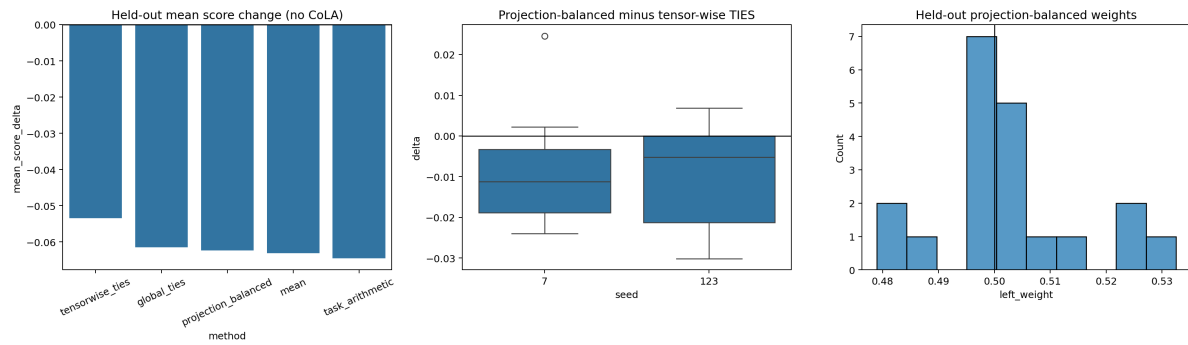


Figure 5. Held-out test of projection-balanced merging. The scalar weights remain close to equal averaging and do not improve tensor-wise TIES.